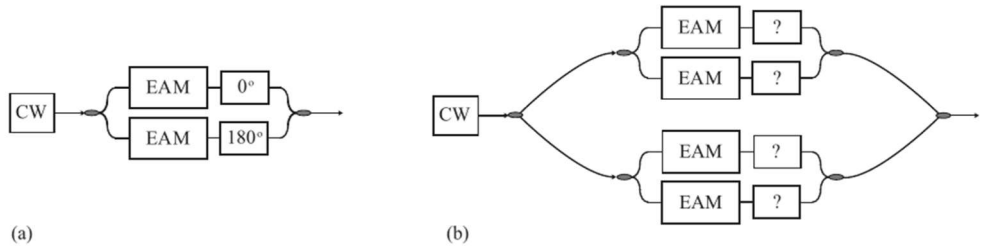
Advanced Optical Fiber Communications

Problem 20: Transmitters

The transmitter in Fig. a) can be used to generate both on-off keying (OOK) and binary phase-shift keying (BPSK). An incoming continuous wave (CW) signal is split equally into two branches. In each branch there is an electro-absorption modulator (EAM) which can turn the light on and off, and a phase shifter which provides a constant phase shift. The signals in the two branches are then recombined at the transmitter output.



a) Explain in words and illustrate your explanation using constellation diagrams how the transmitter can be used to generate both OOK and BPSK.

b) In Fig. 1b, two devices such as the one in Fig. la are connected in parallel. What should the phase shifts in the different branches be to make it possible to generate quadrature phase-shift keying (QPSK) symbols? Motivate your answer. Notice that the value of the phase shifter is an "effective value" including any possible phase shift in the splitters/combiners. (It is only necessary to state one of the different possible combinations.)

Problem 21: Long Range Transmission

In a digital coherent receiver, chromatic dispersion (CD) can be compensated after detection with a finite-impulse response filter whose number of taps depends on the transmission distance, the dispersion parameter, and the symbol rate. You are designing a coherent receiver for a 40 Gbit/s polarization-multiplexed QPSK system () = 1550 nm) and plan to use a rule-of-thumb which says that four filter taps are required per 1000 ps/nm of accumulated CD.

a) How many filter taps are required to compensate for the CD in the link shown, which consists of 10 identical sections with single-mode fiber?

b) The link in has now a DCF module after each SMF section, but the DCF is a bit too short and there is a residual CD after transmission. How many taps are required for the compensation?

(c) How does the impact of dispersion (and therefore also the number of taps in the compensation filter) scale with the symbol rate? Use your answer to estimate the number of filter taps needed to compensate for 1000 ps/nm for a 28 Gbaud polarization-multiplexed QPSK signal.

